DASE7506 MP2
Post-deadline research closeout

Student 3036900957 | Research archive only | Not a new course submission

POST-DEADLINE RESEARCH ARCHIVE: V3 FORMAL TEST AND SCORE UPDATE NOT PERFORMED.

The formal deadline was missed. No V3 official test was run, no new score Issue was posted, and no V3 score update is claimed. The earlier V1 Issue #294 remains the submitted result: 348/467 (74.5182%). This historical test score does not evaluate the V3 archive model.

No locked V3 candidate improved either overall or hard validation counts over the preserved V2 fallback. The historical V2 checkpoint is retained byte-for-byte; no accepted replacement is claimed.

The retained research model is deepsets_regularized_s0. Its saved offline validation outcome is 740/798, hard 297/355. Independent review and a fresh-environment validation reproduction passed before archive freeze. These checks establish reproducibility; they neither reopen submission nor establish a test score or rank.

Evidence	Actual scope
Complete V3 runs	35 common-three-fold configurations
Preserved V2 runs	19 historical configurations
Fixed fusions	4 rejected comparisons
Incomplete campaign jobs	4 excluded from complete-CV tables
V3 formal test / update	NOT PERFORMED

Four added-seed jobs were pruned without complete CV. Preserved attempt records include an elapsed wall duration of 186401.74966692924 seconds across 2026-10-30 to 2026-11-01, with long gaps in monitoring; this is not evidence of continuous CPU work or uninterrupted supervision. The underlying cause of the gaps is unestablished. Original nominal attempt and absolute campaign deadlines were not reset; resumption stopped overdue work. The formal closure occurred on 2026-11-02 after both the research budget and course score deadline. The reserved delivery buffer was missed, and no timely V3 submission is claimed.

Deadline: 2026-10-31 16:00 UTC (2026-11-01 00:00 UTC+8). Research closure after expiry is documented, not a claim that every planned experiment finished within the original budget.

1. Representation and controlled comparisons

Protocol: 4755 training questions, 798 validation questions and 467 official test questions (V3 test unopened). The common 3 outer folds each hold out 1585; the remaining 3170 split into 2536 inner-fit and 634 stopping questions. Fresh outer refits use all 3170 with inner-selected epochs. All four candidates receive correctness labels; alternate correct answers stay positive. Hard questions have at least two distinct parseable answer groups. OOF-based architecture selection makes reported CV and paired intervals selection-optimistic.

Four candidate answers share an official answer parser and one question-level decision. Relation inputs contain 1,024 embedding coordinates, their 1,024 absolute deviations from the candidate mean, and 16 parsing/vote/text statistics. Reduced inputs omit the deviation block. Features exclude IDs and references. Scalers fit only the fitting partition.

The selector masks unparseable answers, groups equal parsed values and uses group log-sum-exp minus alpha times log group size, at tau=1. Deterministic ties return an original candidate index; all-unparseable rows return index0 with confidence zero. TabM averages member answer-group probabilities, not logits.

Architecture / control	Implementation tested
MLP / ResMLP	Shared candidate network / residual transformations
Deep Sets / Attention [3,4]	Permutation-compatible candidate context, without positional IDs
TabM [1]	Shared matrices with member adapters; average member losses
RealMLP-inspired [2]	Input scale, NTP, SELU, zero head and robust preprocessing controls
Muon / EMA [5]	Hidden-matrix Muon plus auxiliary AdamW; same-trajectory post-step EMA

R05 matched TabM recipe: embedding+statistics (1040 inputs) versus full relation (2064) scores 1668/2114 versus 1606/2114 hard questions: +62 (+2.93 pp; saved paired 95% interval [1.51, 4.40] pp), with fold 0/1/2 gains +45/+11/+6; all-question gain +62. This is not a pure feature-information effect: input width changes first-layer capacity and random initialization even at the same fixed seed. The interval is conditional on selected OOF predictions and retains search optimism.

No-fit train reference	All correct	Hard correct	Hard mean +/- SD
fixed0	3584/4755	1078/2114	50.992 +/- 2.165
valid_answer_majority	4054/4755	1413/2114	66.840 +/- 0.642

Fixed0 always returns original index0. Majority ignores None, groups equal official parsed values, breaks ties by earliest original index and returns0 if all are unparseable. Their fold statistics are descriptive partitions, not three fits.

2. Complete V3 training experiments (1/2)

All values below are training OOF results on common question folds. Hard fold mean and sample SD are percentages; fit seconds sum the recorded inner/outer optimization cost. Full configuration IDs are immutable identifiers, not encodings of learning rates.

Experiment ID	All correct	Hard correct	Hard mean +/- SD	Fit s
v3_tabm_embedding_stats_ema_s0	4328/4755	1687/2114	79.80 +/- 1.94	231.3
v3_attention_ema_s0	4326/4755	1685/2114	79.71 +/- 1.27	266.3
v3_attention_pairwise_s0	4325/4755	1684/2114	79.66 +/- 0.65	210.5
v3_attention_wd01_s0	4324/4755	1683/2114	79.61 +/- 0.80	80.4
v3_attention_adamw_s0	4323/4755	1682/2114	79.56 +/- 0.75	181.7
v3_realmlp_embedding_stats_drop01_s0	4323/4755	1682/2114	79.56 +/- 0.79	143.2
v3_attention_width256_s0	4318/4755	1677/2114	79.33 +/- 1.08	186.7
v3_attention_drop05_s0	4314/4755	1673/2114	79.14 +/- 0.57	116.6
v3_attention_listwise_s0	4313/4755	1672/2114	79.09 +/- 1.33	170.8
v3_attention_small_s0	4311/4755	1670/2114	79.00 +/- 0.65	111.1
v3_attention_bce_s0	4309/4755	1668/2114	78.90 +/- 0.85	238.0
v3_tabm_k4_embedding_stats_s0	4309/4755	1668/2114	78.90 +/- 2.08	279.2
v3_attention_muon_s0	4307/4755	1666/2114	78.81 +/- 0.54	2288.8
v3_realmlp_robust_embedding_stats_s0	4306/4755	1665/2114	78.76 +/- 1.14	71.6
v3_deepsets_ema_s0	4306/4755	1665/2114	78.76 +/- 1.44	123.7
v3_attention_group_alpha05_s0	4304/4755	1663/2114	78.67 +/- 0.48	180.5
v3_tabm_embedding_stats_k8_s0	4304/4755	1663/2114	78.67 +/- 1.56	265.7
v3_realmlp_embedding_stats_ema_s0	4299/4755	1658/2114	78.43 +/- 1.02	167.9

All rows, including losses, rates, preprocessing, decoder, source hashes and numeric asset bytes, remain in the bound final_search/EXPERIMENT_TABLE.md and original per-experiment summaries.

3. Complete V3 training experiments (2/2)

All values below are training OOF results on common question folds. Hard fold mean and sample SD are percentages; fit seconds sum the recorded inner/outer optimization cost. Full configuration IDs are immutable identifiers, not encodings of learning rates.

Experiment ID	All correct	Hard correct	Hard mean +/- SD	Fit s
v3_deepsets_adamw_s0	4298/4755	1657/2114	78.38 +/- 1.32	144.3
v3_resmlp_ema_s0	4266/4755	1625/2114	76.87 +/- 1.00	67.6
v3_resmlp_adamw_s0	4262/4755	1621/2114	76.68 +/- 0.72	47.4
v3_tabm_k8_s0	4260/4755	1619/2114	76.58 +/- 0.73	321.9
v3_tabm_k4_ema_s0	4260/4755	1619/2114	76.58 +/- 0.76	579.3
v3_realmlp_robust_all_s0	4252/4755	1611/2114	76.21 +/- 0.71	73.9
v3_realmlp_robust_lr0003_s0	4251/4755	1610/2114	76.16 +/- 0.73	95.4
v3_tabm_k4_lr0003_s0	4251/4755	1610/2114	76.16 +/- 0.57	471.8
v3_realmlp_robust_width256_s0	4248/4755	1607/2114	76.02 +/- 1.03	111.4
v3_tabm_k4_s0	4247/4755	1606/2114	75.97 +/- 0.93	157.0
v3_tabm_k4_drop03_s0	4246/4755	1605/2114	75.92 +/- 0.73	185.0
v3_mlp_ema_s0	4242/4755	1601/2114	75.73 +/- 1.15	73.3
v3_realmlp_robust_stats_s0	4240/4755	1599/2114	75.64 +/- 1.14	50.1
v3_mlp_muon001_s0	4239/4755	1598/2114	75.59 +/- 0.48	1079.0
v3_realmlp_standard_s0	4238/4755	1597/2114	75.54 +/- 1.35	50.2
v3_mlp_adamw_s0	4233/4755	1592/2114	75.31 +/- 1.04	49.5
v3_mlp_muon002_s0	4230/4755	1589/2114	75.17 +/- 0.14	790.0

Recorded completed-V3 fit time totals 9660.9 s. Recorded process-attempt wall time totals 208968.9 s; 0 attempts lack that field. These clocks have different scopes and are not added. Wall time includes elapsed gaps and startup; it does not establish continuous CPU execution or continuous monitoring.

4. Preserved historical V2 experiments

All values below are training OOF results on common question folds. Hard fold mean and sample SD are percentages; fit seconds sum the recorded inner/outer optimization cost. Full configuration IDs are immutable identifiers, not encodings of learning rates.

Experiment ID	All correct	Hard correct	Hard mean +/- SD	Fit s
attention_regularized_s0	4328/4755	1687/2114	79.80 +/- 0.72	133.3
attention_hard_ablation	4321/4755	1680/2114	79.47 +/- 0.89	133.3
attention_regularized_s1	4319/4755	1678/2114	79.38 +/- 0.72	1098.4
deepsets_regularized_s1	4313/4755	1672/2114	79.09 +/- 1.03	59.9
attention_regularized_s2	4312/4755	1671/2114	79.04 +/- 1.39	60.2
attention_group_s0	4309/4755	1668/2114	78.90 +/- 1.85	41.7
deepsets_regularized_s2	4308/4755	1667/2114	78.85 +/- 1.09	59.7
deepsets_no_context	4296/4755	1655/2114	78.29 +/- 0.87	86.9
deepsets_regularized_s0	4295/4755	1654/2114	78.24 +/- 0.96	117.8
deepsets_group_s0	4289/4755	1648/2114	77.96 +/- 1.14	34.3
mlp_listwise_s0	4253/4755	1612/2114	76.25 +/- 1.25	25.9
mlp_capacity_s0	4244/4755	1603/2114	75.83 +/- 0.84	42.8
linear_group_a0	4242/4755	1601/2114	75.73 +/- 1.21	14.3
linear_group_hard	4242/4755	1601/2114	75.73 +/- 1.21	13.0
resmlp_group_s0	4241/4755	1600/2114	75.69 +/- 0.69	27.4
mlp_group_s0	4240/4755	1599/2114	75.64 +/- 0.52	32.1
linear_group_a0p5	4236/4755	1595/2114	75.45 +/- 0.61	14.5
linear_group_a1	4229/4755	1588/2114	75.12 +/- 0.48	13.0
linear_original	4213/4755	1576/2114	74.55 +/- 0.65	11.7

Historical V2 recorded fit time: 2020.1 s. Those earlier runs and assets are retained unchanged. They are not additional V3 configurations or newly obtained test results.

5. Decoder, objective and stability evidence

Saved V2 control	Hard correct	Hard mean +/- SD
linear_original	1576/2114	74.551 +/- 0.651
linear_original::mask_max	1578/2114	74.646 +/- 0.811
linear_original::group_max	1578/2114	74.646 +/- 0.811
linear_original::group_lse_a0	1604/2114	75.875 +/- 0.369
linear_original::group_lse_a0.5	1595/2114	75.450 +/- 1.107
linear_original::group_lse_a1	1552/2114	73.416 +/- 0.622
linear_group_a0	1601/2114	75.733 +/- 1.210
linear_group_a0p5	1595/2114	75.450 +/- 0.607
linear_group_a1	1588/2114	75.118 +/- 0.482
linear_group_hard	1601/2114	75.733 +/- 1.210
mlp_group_s0	1599/2114	75.638 +/- 0.515
resmlp_group_s0	1600/2114	75.686 +/- 0.692
deepsets_group_s0	1648/2114	77.956 +/- 1.139
attention_group_s0	1668/2114	78.902 +/- 1.853

The six linear_original rows reuse identical scores to isolate masking/group decoding. Original listwise training includes all candidates in its denominator and all positives in its numerator; it does not relabel alternate correct answers negative. Group CE, listwise, BCE and valid positive-negative softplus pairs are distinct registered objectives. Hard/full reduction requires corresponding regularization accounting.

Completed identical recipe	Seed suffixes	Mean hard %	Across-seed SD
research_v2/attention_regularized_s0	957,958,959	79.41	0.38
research_v2/deepsets_regularized_s0	957,958,959	78.73	0.44

Only genuinely completed identical recipes enter this descriptive seed table. Across-seed sample SD is not a confidence interval or fold SD. The deployment seed remains3036900957. Incomplete V3 repeat seeds cannot establish V3 seed stability.

6. Research rounds, interruptions and limits

Registered round	Jobs	Complete	Best base-seed hard mean %
r01_mlp_optimizer	4	4	75.73
r02_tabm	3	3	76.58
r03_realmlp_preprocess	3	3	76.21
r08_fixed_heterogeneous_fusion	0	0	No complete primary run
r04_attention_optimizer_capacity	4	4	79.71
r05_tabm_optimization_features	3	3	78.90
r06_attention_objectives	4	4	79.66
r07_regularized_interactions	4	4	78.76
r09_attention_regularization_capacity	3	3	79.61
r10_realmlp_resolution	3	3	78.76
r11_tabm_reduced_features_variance	2	2	79.80
r12_reduced_tabm_attention_equal_fusion	0	0	No complete primary run
r14_realmlp_reduced_variance	2	2	79.56
r15_reduced_tabm_ema_attention_equal_fusion	0	0	No complete primary run
r13_fixed_seed_stability	4	0	No complete primary run

Incomplete job	Terminal status	Last recorded reason
v3_tabm_embedding_stats_ema_s1	pruned	attempt_wall_budget
v3_tabm_embedding_stats_ema_s2	pruned	attempt_wall_budget
v3_attention_ema_s1	pruned	absolute_campaign_budget
v3_attention_ema_s2	pruned	Not started: absolute_campaign_budget

Incomplete jobs and partial folds remain preserved and are excluded from whole-CV rankings. A logged fold seed is the fixed base seed plus the fold offset, not an independently selected repeat. Epoch checkpoints preserve model/optimizer/EMA/RNG state; a resumed attempt is not counted as another completed configuration.

Original planning used a 24-hour absolute research clock and per-attempt budgets. Actual monitoring gaps, late closure and deadline failure are disclosed on page1; no claim of uninterrupted 24-hour CPU work or successful budget enforcement is made. Missing timing is not invented. Historical ledger rows without an analysis hash: r01_mlp_optimizer. Their currently verified summaries are bound without rewriting prior records.

7. Negative fixed fusions and paired uncertainty

Fixed probability mixture	Hard correct	Delta vs named matched member	Decision
v3_fusion_attention_tabm_s0	1667/2114	-15 vs v3_attention_adamw_s0	Rejected
v3_fusion_attention_realmlp_s0	1670/2114	-12 vs v3_attention_adamw_s0	Rejected
v3_fusion_reduced_tabm_attention_ema_s0	1680/2114	-5 vs v3_attention_ema_s0	Rejected
v3_fusion_reduced_tabm_ema_attention_ema_s0	1687/2114	0 vs v3_tabm_embedding_stats_ema_s0	Rejected

v3_fusion_reduced_tabm_ema_attention_ema_s0: +2 hard questions versus v3_attention_ema_s0 (1685 correct), but +0 versus the matched v3_tabm_embedding_stats_ema_s0 (1687 correct); rejected.

Member training costs are counted once in the campaign. OOF mixing is a deterministic diagnostic, with no new optimization. Formal fresh-process latency was not measured for rejected mixtures.

Each fusion followed saved reciprocal-error/oracle evidence and one registered equal-probability mixture. All four were rejected. No result is rescued by a post-hoc weight, temperature, member, seed or routing sweep. Passing a complementarity gate alone did not guarantee an improvement.

Base-seed round representative	Hard delta pp	Saved paired 95% interval
r01_mlp_optimizer	-4.07	[-5.58, -2.55]
r02_tabm	-3.22	[-4.59, -1.80]
r03_realmlp_preprocess	-3.60	[-5.01, -2.13]
r04_attention_optimizer_capacity	-0.09	[-0.76, 0.57]
r05_tabm_optimization_features	-0.90	[-1.80, 0.00]
r06_attention_objectives	-0.14	[-1.23, 0.99]
r07_regularized_interactions	-1.04	[-2.03, 0.00]
r09_attention_regularization_capacity	-0.19	[-0.57, 0.19]
r10_realmlp_resolution	-1.04	[-2.18, 0.09]
r11_tabm_reduced_features_variance	0.00	[-0.99, 0.99]
r14_realmlp_reduced_variance	-0.24	[-1.18, 0.71]

Round representatives are selected from completed train OOF results and compared with preserved V2 Attention predictions. Saved paired intervals are conditional on those predictions and retain search-selection optimism. They are not test intervals, proof of a causal architecture effect, or a ranking claim.

8. Offline confirmation and clean reproduction

After search closure, at most two fixed-seed finalists were locked for validation. The predeclared confirmation/fallback rule selected one archive checkpoint. Validation had already informed earlier work; this is not an untouched-holdout estimate. No subsequent test was opened.

Locked candidate / fallback	All validation	Hard validation
Historical V2 fallback	740/798	297/355
v3_tabm_embedding_stats_ema_s0	735/798	292/355
v3_attention_ema_s0	736/798	293/355

No locked V3 candidate improved either overall or hard validation counts over the preserved V2 fallback. The historical V2 checkpoint is retained byte-for-byte; no accepted replacement is claimed.

Component	Family	Seed	Epochs	Preprocess	Weights
deepsets_regularized_s0	deepsets	3036900957	9	standard	live

Fresh-environment reproduction retained identical saved choices and official validation score: 740/798. Prediction plus scoring took 2.280391 s; frozen inference assets total 963670 uncompressed bytes. These are validation reproduction measurements, not V3 test latency or acceptance by the course.

Exact configuration, preprocessing, decoder, seed/epoch aggregation and EMA/live weight choice are bound in FREEZE.training_recipes and the original final_fit records. Independent contract/export checks and clean-environment receipts must pass before this report can be generated.

Chosen-event diagnostic	Brier	ECE10	AURC
Train OOF raw	0.0627	0.0191	0.0131
Train OOF mapped (in-sample)	0.0618	0.0053	0.0131
Used validation raw	0.0541	0.0161	0.0101
Used validation mapped	0.0548	0.0201	0.0101

Confidence is the probability of the actually chosen answer group. The fixed C=1 logistic map fits train OOF only; its fitted-OOF metrics are IN-SAMPLE, and validation was already used for confirmation. Ten equal-width ECE bins retain counts; AURC summarizes saved risk-coverage. The diagnostic changes no choices and is not loaded by formal prediction.

9. Reproduction, provenance and honest limits

Archive source commit: eee478b4959e555b6ec8c902b6db8b69f31df508. Model: code/solution_v3.py; assets: artifacts/research_v3/final_selected. For fresh reproduction, use a separate clone checked out at this exact source commit, before the later archive FREEZE metadata, and prepare the fixed official data with code/tools/fetch_official.py; the isolated evaluation checkout receives validation files only. Verify retained source/asset hashes and package versions; then run python scripts/reproduce_selected_v3.py --attempt independent_validation. Its preflight intentionally refuses the current frozen archive directory. Preserve that directory and its guards; do not delete FREEZE or reopen research, test or score updates.

Training regeneration uses the committed configurations and common_folds.json through code/train_v3.py, including fold-local scaling, nested stopping and fresh refits. Existing closed-search guards remain active. Reproduction of archived predictions is separate from reopening model research.

AI assistance was substantive: source inspection, implementation, experiment orchestration, diagnostics, report preparation and internal AI review. Actual logs and hashes support numerical claims; AI review is not instructor approval or a claim of independent human understanding. Author commits, licenses and adaptation details are retained in MODELS_SOURCES.md, OPTIMIZER_SOURCES.md and model-source-evidence.json.

Primary method references

[1] Gorishniy et al. TabM: Advancing Tabular Deep Learning with Parameter-Efficient Ensembling. ICLR 2025. https://arxiv.org/abs/2410.24210

[2] Holzmuller et al. Better by Default: Strong Pre-Tuned MLPs and Boosted Trees on Tabular Data. NeurIPS 2024. https://arxiv.org/abs/2407.04491

[3] Zaheer et al. Deep Sets. NeurIPS 2017. https://arxiv.org/abs/1703.06114

[4] Vaswani et al. Attention Is All You Need. NeurIPS 2017. https://arxiv.org/abs/1706.03762

[5] Gorishniy et al. Benchmarking Optimizers for MLPs in Tabular Deep Learning. 2026 preprint, inspected v1 (Muon/EMA motivation). https://arxiv.org/abs/2604.15297v1

Limitations: missed formal deadline; monitoring gaps; four incomplete V3 seed repeats; selection-optimistic CV/bootstrap; validation reuse; known historical test aggregate; compact task adaptations rather than full source-paper benchmarks; negative fusions; and no new V3 test result. No training or validation statistic establishes a leaderboard position.

Peer review is handled through the instructor's actual released workflow as a separate task. This report neither marks it complete nor postpones it by assumption. Consult its independent current evidence.

Evidence index: final_search/EXPERIMENT_TABLE.md; round_decisions.jsonl; campaign_queue.json; fixed_fusion_analysis_complete.json; deterministic_train_references; R05 paired-and-slices; SELECTED_MODEL/FREEZE/REVIEW_RESULT/REPRODUCTION_RESULT; FORMAL_SUBMISSION_BLOCKED/RESEARCH_CLOSEOUT. All are hash-bound in closeout_inputs.json.

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